COURSE 2 · SECTION 9 · EXERCISE

Field ppm by seat ring revision

Section 9 (videos S09.1 and S09.2) · Validation pass rate + field ppm

Start: Section10_Start.pbix or your Engxtech_S09_2.pbix; save a copy as Engxtech_S09_Exercise.pbix

Time: about 25 min

Data: the course data folder

DESIGN QUESTION

Does the field data show a difference in seat ring wear between revs A, B and C, and how sure can we be about rev C?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

TestResults has one row per validation test lot (this is a validation test pass rate, not production first-pass yield). Builds gives units built per product and month with the seat ring rev fitted; WarrantyClaims has one row per claim. The video linked both to the small tables SeatRingRevs and BuildMonths and to Products, and added the Matured, Within90 and IsSeatRingWear columns.

Your tasks

1. In your exercise copy, read **Validation Pass Rate** on the Validation & field page.
2. Read the ppm table by SeatRingRevs[SeatRingRev]: **Wear Failures 90d**, **Matured Units**, **Field ppm**, **CI Low ppm**, **CI High ppm**. Which build months count as matured?
3. New measures in _Measures: **Wear Failures 90d (all months)** and **Units Built (all months)**, the same as the video's measures but without the Matured filter.
4. New measure **ppm (all months, not valid)** from those two. Read it for rev C and explain why it is lower.
5. Write one sentence about rev C that a design reviewer could quote. It must not say that rev C fixed the problem.

Check your answer

Check	Expected
Validation test pass rate	**90.9 % (149/164)**
Matured build months	**Mar 2025 to Jun 2026**
Rev A	**2255 ppm (8/3548); 95 % CI 973-4443 ppm**
Rev B	**2892 ppm (81/28007); 95 % CI 2297-3595 ppm**
Rev C	**1710 ppm (4/2339); 95 % CI 466-4379 ppm**
Rev C incl. unmatured months (not a valid rate)	**850 ppm (8/9407)**

Hint. Rev C has one matured build month and 4 failures. Its interval includes rev B's rate and overlaps rev A, so the data cannot yet tell rev C apart from rev B or rev A: an early signal, not proof that the change worked.

Stuck? The worked solution with the full DAX and M is in Solutions/S09_Solution.pdf.